SoK: Evaluating LLM and Agent Side-Channel Defenses From Evidence Audits to Real-Agent Measurements

***Abstract*—Encryption conceals content while exposing the structure of agent workflows. We systematize 82 study families through 39 source-bound result records, separating the observer, execution boundary and evaluation conditions that determine whether defense evidence transfers. The audit exposes three recurring mismatches: protection attached to the wrong receiving view, utility attached to a different state boundary, and cost attached to a different mechanism or attacker. Experiments on 690 real-agent runs turn these distinctions into measurable consequences. Customer-grouped cross-fitting shows approximately 30–35 percentage-point paired reductions in balanced accuracy under joint shaping in the standard-prompt arm, while a write-only ablation locates the dominant signal beyond a rules prompt's read-order constraints. The prompt supplies a behavioral control. Live tool measurements and stronger attacker comparisons expose timing leakage and substantial resource costs; the absolute security criterion remains unconfirmed. Classical traffic-shaping controls locate these results within a workload-specific cost comparison. The contribution is an auditable connection from published evidence to mechanism-level findings and deployment decisions.**

1. Introduction

Encryption conceals message content, yet LLM engines and agents expose packet sizes, cache timing, visited domains and tool sequences [1], [2], [3], [4], [5], [6]. A defense result applies to a particular receiver, execution boundary and workload. Reusing a headline score across a frontend, dispatcher or tool provider can change both its protected signal and its cost object.

Our SoK uses the *evaluated result record* to make these transfers inspectable. Separating Dangol's stored-capture and live-protocol results prevents subtracting unmatched scores; separating Early Bird's KV-prefix and semantic-cache mechanisms prevents pricing one with the other's overhead. Combining signal origin with observer access and threat polarity explains why read-only privacy cover and state-changing bot evasion have different utility consequences. These are systematization findings: they emerge from binding evidence across records, rather than from proposing a new defense component.

Contributions.

- **A populated evidence framework.** A complete map of 58 core and 24 adjacent families, 39 source-bound records and a nine-item audit of 20 mitigation families. The framework identifies which comparisons are compatible, incompatible or unresolved.
- **Three reusable comparison judgments.** Bind the final receiving projection; bind execution to the protected state; bind adaptive evaluation, utility and cost to the same operating point. Only 3/17 applicable families bind utility and 2/11 bind matched adaptation. The audit identifies the missing evidence needed for reuse.
- **Mechanism and cost evidence on a common agent–tool chain.** Controlled streaming/state tests and 690 real-agent runs show why emitter counts, state-changing calls and defended-data fitting matter. Customer-grouped cross-fitting supplies the main within-cohort summary; the original frozen-calibration split remains a separate reference. Live tool timing, stronger attackers and classical traffic-shaping adaptations test the boundaries of the measured reductions.

Shadow isolation, decoys, constant-rate tunnels and bucket padding have prior art; our contribution is their evidence-bound comparison. In particular, a high-overhead operating point is useful for diagnosing a protection boundary without being deployment-ready. Table 1 states the answers to the four research questions and the evidence supporting each.

Scope of claims. The corpus supports source-bound comparison judgments; the experiments support paired, workload-specific leakage and cost measurements. Original cross-fitting measures length/order; E13 additionally measures local arrival timing. Neither certifies an absolute bound against arbitrary attackers. The principal model cohort uses one retail environment and one mutable model alias. Preserved logical transcripts do not establish unchanged reward under timing-dependent LLM replanning.

Figure 1 puts trace and action paths on the same agent–tool chain. Classical data processing motivates the receiving-view check; it is not a new theorem. A detailed decision flow is retained with the supplementary methods.

2. Scope and Systematization Method

2.1. Research questions, eligibility and threat polarity

We organize the systematization around four questions: which hidden targets, threat polarities and auxiliary observables appear across LLM and agent systems (RQ1); how

TABLE 1. RESEARCH QUESTIONS, SYSTEMATIZATION FINDINGS AND EVIDENCE. EXPERIMENTAL FOLLOW-UPS TEST THE COMPARISON RULES; THE SOK CONTRIBUTION IS THE RECORD-LEVEL SYNTHESIS.

Question	Finding	Evidence in the main paper
RQ1: targets and observables	A common signal can serve confidentiality, attribution, evasion or authorized diagnostics; polarity changes the protected interest.	Complete core-family map (§4, Table 6).
RQ2: origin versus access	Origin A/B does not determine receiver access. Upstream protection need not survive a plaintext-aware downstream emitter; read cover does not imply safe writes.	Record lineage and three literature cases (§3–5); E5–E7.
RQ3: portability	A task holdout does not establish topology, hardware or model transfer; mixed configurations can preserve shortcuts.	Generator-specific holdouts (§7); customer-grouped analysis (§8.5).
RQ4: protection, state and cost	Evaluate leakage, logical state preservation and resource cost together. Full LLM utility under timing-dependent replanning is not established.	Twenty-family audit; E9–E14 leakage/state/cost contrasts (§8).

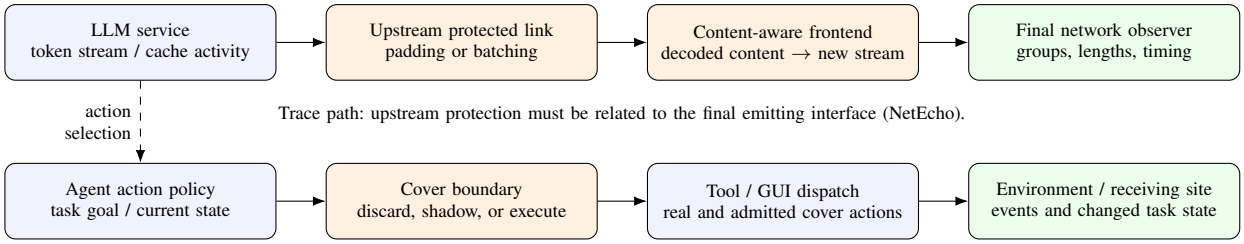


Figure 1. Observation and execution boundaries. E5 tests streaming, E6 state scope, E7 encrypted composition, E9 frozen-calibration real-agent trajectories, and E13 live tool timing with complete tunnel controls.

signal generation origin relates to receiver observation permissions (RQ2); which holdouts establish portability across tasks, configurations and hardware (RQ3); and which evidence binds workflow protection, logical state preservation and resource cost, distinguishing these from full end-to-end planner utility (RQ4).

Core inclusion requires five conjunctive criteria: (C1) LLM or agent execution, (C2) a hidden target property, (C3) an execution-dependent auxiliary observable rather than ordinary output semantics alone, (C4) specified observer permissions, and (C5) an empirical inference evaluation. Mixed-feature papers qualify only if they isolate an auxiliary-only C3 component; thus coding fingerprints, WildAuth, ZeroPrint, endpoint-content fingerprints and StepFinder remain adjacent boundary comparators [7], [8], [9], [10], [11]. Table 2 states the observable eligibility rules.

We use four narrative groupings of objectives and observer roles, not mutually exclusive classes. An unauthorized identity inference can be both attribution and a confidentiality concern. Each record separately specifies the inference target, observer, authorization relation and protected interest: **Type I** concerns confidentiality/privacy; **Type II** concerns model, harness or identity attribution; **Type III** concerns evasion of a platform's human–agent detector; **Type IV** concerns authorized operator diagnostics. Types can overlap. WRA protects research intent (I); UI Traces attributes agents (II); AHB evades a defensive detector (III); ClawGuard diagnoses loops (IV) [6], [5], [12], [13]. These roles are mapped per core family, alongside generating and receiving layers, in Table 6.

TABLE 2. OBSERVABLE-BASED ELIGIBILITY. A HIDDEN TARGET ALONE DOES NOT MAKE A SEMANTIC ATTACK A SIDE CHANNEL.

Observed input	Decision and required distinction
Length, timing, cache response	Core when measured execution dependence supports hidden-target inference.
Own-site actions	Core attribution with site permission; no automatic access to other sites or private tool logs.
Reply states the secret	Direct disclosure; adjacent, not auxiliary-channel evidence.
Reply style/content identifies a user	Adjacent unless a separately evaluated auxiliary mechanism is established. WildAuth stays a mixed-view comparator.
Output-filter membership oracle	Can be core: the filter's execution, not ordinary semantic output inference, mediates the signal.
Privileged loop/fault logs	Separate diagnostic component when C1–C5 hold; does not increase confidentiality evidence.

2.2. Corpus construction and reconciliation

We construct a purposive corpus through 40 seeds, venue-targeted proceedings screening and one-hop citation snowballing, supplemented by native database and bulk

keyword searches. The completed supplementary search returned 615 occurrences and 565 unique records; an enriched audit of 204 exclusions rerouted four records, yielding 15 new core and seven adjacent families. The resulting collection contains 58 core and 24 adjacent families; Table 6 maps every core family to its signal and observer layers and threat roles. The query set overlaps 17/24 selected seed papers and 25/60 earlier indexed families, motivating complementary discovery routes; full queries, exclusion accounting and version reconciliation are in Supplement S1/S10.

2.3. Coding procedure and consistency

AI assistance produced initial schema-guided field drafts. Authors checked and corrected all 39 result records and the 180 archived audit cells against source section/table/figure/paragraph locators, and reviewed all 82 family classifications using titles, abstracts and experimental sections. These version-bound locators make the evidence inspectable. Agreement compares the archived initial extraction and subsequent author-verification columns: 93/100 screening decisions ($\kappa = 0.886$), 134/144 audit cells ($\kappa = 0.876$) and 141/150 pair judgments ($\kappa = 0.916$). These are explicitly non-independent extraction–verification statistics: the author pass corrects the initial AI-assisted draft. No independent human inter-rater κ is available. The 36 cells from four added families are author-verified but outside that agreement sample. A subsequent source-linked AI-assisted reconciliation updates three displayed statuses while preserving the historical columns and agreement values (S8/S18).

SoK Takeaway 1 (Map evidence to the receiving deployment). A family-level label does not identify which signal, observer permission or protected objective a result supports. The complete core map separates signal origin from receiving access; the result ledger then binds each comparison to a source version and operating point.

TABLE 3. A SINGLE RESULT RECORD. ONLY FIELDS RELEVANT TO THE REQUESTED COMPARISON MUST ALIGN; UNAVAILABLE REQUIRED FIELDS REMAIN UNRESOLVED.

Record field	Required meaning
Source	Family/result ID, version, source locator, reading scope and hash.
Target	Hidden property, inference goal and protection relation: confidentiality, attribution, human/agent distinction or authorized diagnosis.
Receiver	Actual permission and released representation, including session/document linkage and any auxiliary inputs.
Intervention	Where protection ends; whether edits affect stored traces, transmission or executed actions.
Attacker	Fit/calibration condition, test population, training access and adaptation budget.
Workload	Task, system, topology/hardware and grouping/holdout.
Outcomes	Inference metric / denominator; objective or proxy utility; applicable cost object; uncertainty and field states.

TABLE 4. COMPARISON RULES. A KNOWN MISMATCH AND A MISSING BINDING HAVE DIFFERENT CONSEQUENCES.

Requirement	Condition and resulting license
Descriptive comparison	Align target, receiver permission, metric meaning and cost object. Different systems may be described together; token payload and wire bytes cannot be ranked as the same cost.
Paired effect	Compare the same defended test units and outcome. A known test-unit mismatch makes that subtraction incompatible; an unverified fit binding leaves the effect unresolved.
Deployment transfer	Establish a bridge for the changed receiver, intervention, workload, topology or hardware. A demonstrated violation blocks the proposed transfer; an untested bridge leaves it unresolved. Permit only the range and endpoints actually covered.

3. Result Records, Comparability Rules and Validity

The atomic unit of our repository is one *evaluated result record*, rather than a whole paper or a single headline score. A result record binds a source version and locator to seven structured fields: Source, Target (and threat polarity Types I–IV), Receiver projection, Intervention boundary, Attacker training/calibration (D_0 vs. D_2), Workload/holdout, and Outcomes (inference metric, task/component utility, and cost object; Table 3). A study family contributes multiple records whenever it evaluates distinct execution modes (e.g., stored capture vs. live loopback in Dangol [14]), distinct mechanisms (e.g., K-prefix KV simulation vs. GPTCache semantic de-identification in Early Bird [3]), or separately scoped performance benchmarks (e.g., SafeKV [15] and OTRO [16]).

Indexing schema and comparison contracts. The mnemonic SCOPE (Secret/target, Channel, Observer, Pipeline, Exploitation goal) names the target-and-channel columns of our index, paired with evaluation provenance. Generating layers span hardware (P1), serving (P2), transport (P3), application/API (P4), orchestration (P5), tools/MCP (P6), environment (P7) and multi-agent communication (P8). We apply three comparison contracts (Table 4): a comparison is *compatible* when required fields align; *incompatible* when a positive mechanism or boundary conflict is established (such as substituting token-payload ratios for wire bytes or offline timestamp edits for live action delays); and *unresolved* when a required binding is unreported (NR-R), unavailable (NR-A) or unverified (NR-V). Table 5 illustrates a complete record-to-judgment lineage.

Rule-application consistency check. Twenty-four author-constructed cases exercise Compatible/Incompatible/Unresolved rules. Their construction-time agreement checks implementation consistency; they do not estimate decision accuracy on external cases. Supplement S7 preserves every case and outcome. The seven mechanism checks recorded before E5–E7 have the same engineering role, rather than serving as independent validation of the

TABLE 5. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	Dangol workflow / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [14].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about 13× bytes and 11× latency in that condition.
Unresolved fields	Objective terminal task utility NR-R; precise paired live-refit comparison NR-V.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

framework.

4. Corpus Map Across 82 Study Families

4.1. Origins, receiving projections and threat polarities

Across all 58 core families in our unified repository, 38 study inherited runtime, serving or hardware channels (Origin B) and 20 study agent-workflow channels arising from orchestration, tool dispatch, browser navigation and multi-agent protocols (Origin A; Table 6). Signal generation origin is distinct from both receiver location and threat polarity. The full map makes these distinctions visible for each core family.

Cross-origin synthesis. Origin B includes token/packet lengths, cache timing and co-resident hardware channels; Origin A includes visited domains, tool dispatch and browser actions. Receiver permissions cut across these origins: a remote network observer, a same-site script and a privileged diagnostic hook expose different views even when they share an attribution target. Thus Origin A/B cannot substitute for Types I–IV or observer permissions. WRA's read-only privacy decoys, AHB's stateful evasion and SplitDecoy's exposed return path illustrate transfers that fail when these axes are collapsed [6], [12], [29]. The map retains every core family rather than using these examples as a substitute for coverage.

4.2. Relationship to prior systematizations

Prior surveys establish compound-system threat models and confidential-computing boundaries. Banerjee et al. categorize cross-layer compound AI threats under MITRE tactics; Forough et al. distinguish TEE isolation boundaries from side-channel resistance and note that inference protections need not cover agent memory or action coordination [64], [65]. Kotevska et al. propose a six-element privacy contract (asset, observer, channel, permitted disclosure, guarantee, evidence) with composition conditions framed as formal conjectures [66], while classical website-fingerprinting works establish bandwidth/latency bounds for transport shaping (Tamaraw, WTF-PAD, FRONT, Walkie-Talkie) [67], [68], [69], [70]. Our distinct contribution is the populated 82-family / 39-record repository, the nine-item comparability audit, and mechanism checks recorded before the experiments (S12) for the transfers it flags, using loopback TLS replays and a pinned stateful tool simulator. Website-fingerprinting defenses already show that per-packet padding leaves counts and that constant-shape shaping is expensive [67], [68]. E5 measures how that known property invalidates a specific cost transfer for LLM token streams, without claiming it as new.

5. Three Literature Case Studies

Record-level synthesis resolves three deployment decisions: whether upstream shaping survives another workflow hop, whether cover preserves the relevant state, and whether adaptive evaluation and cost describe the same mechanism. The following cases provide the source-local evidence for each judgment.

5.1. Case 1: bind the final target-bearing emitter

The downstream re-emission boundary. When an upstream LLM API applies record padding or token batching, does that protection cover the user-facing connection after an application frontend? NetEcho demonstrates that a content-aware frontend that consumes decoded upstream tokens and re-serializes responses into new chunks (Trace A token counts per group; Trace B character lengths) restores topic and conversation distinguishability [60]. SplitDecoy establishes the same boundary on a backward return path: even when forward split-model activations are obfuscated with decoys, the returned backward gradient exposes nonzero row support on every tested frame across nine seeds [29, Sections 5.1–5.3].

The standard data processing inequality applies when $S \to Z \to Y$ is a Markov chain: target-oblivious post-processing cannot increase $I(S;Z)$ [71, Theorem 2.8.1]. A plaintext-aware frontend or unprotected backward computation changes that premise. The systematization insight is identifying which published records cross this boundary.

5.2. Case 2: resolve the WRA versus AHB cover asymmetry

Two axes of asymmetry: threat polarity and action risk. A comparison that groups WRA's decoy requests [6] and AHB's humanized actions [12] together as "cover traffic" can create an apparent paradox: WRA reports a

TABLE 6. COMPLETE MAP OF THE 58 CORE STUDY FAMILIES. EACH FAMILY APPEARS ONCE; MULTIPLE LAYER/TYPE LABELS DESCRIBE DIFFERENT COMPONENTS OR OBJECTIVES WITHIN THE FAMILY.

Core family	$G \to O$	Type	Core family	$G \to O$	Type
ArchSnoop [17]	$1 \to 1$	II	PromptPeek [2]	$2 \to 4$	I
Behind Bars [18]	$1 \to 1$	I	Remote keylogging [1]	$2 \to 3$	I
Edge EM leakage [19]	$1 \to 1$	I	Remote timing [20]	$2 \to 3$	I
Energon [21]	$1 \to 1$	II	Routing metadata [22]	$2 \to 4$	I
I Know What You Said [23]	$1 \to 1$	I	SafeKV [15]	$2 \to 4$	I
MoEcho [24]	$1 \to 1$	I	Speculative decoding [25]	$2 \to 3$	I
PAGE404 [26]	$1 \to 1$	I	SpliceLeak [27]	$2 \to 4$	I
Spill the Beans [28]	$1 \to 1$	I	SplitDecoy [29]	$2 \to 2$	I
Detokenization Leaks [30]	$1, 2 \to 1$	I	Time Will Tell [31]	$2 \to 4$	I
LLM-serving RCA [32]	$1, 2 \to 2$	IV	Whisper Leak [33]	$2 \to 3$	I
LLM-WFIN [34]	$1, 2 \to 1, 3$	I	MIRAGE [35]	$2, 5 \to 2$	IV
OTRO [16]	$1, 2 \to 1$	I	Cloud traffic benchmark [36]	$2, 5, 8 \to 3$	I, II
Shape and Substance [37]	$1, 2 \to 1$	I	Web-agent unmasking [38]	$3, 7 \to 3, 7$	II, III
SparLeak [39]	$1, 2 \to 1$	I	LumiMAS [40]	$4, 5 \to 4, 6$	IV
SparSEEty [41]	$1, 2 \to 1$	I	AgentPrint [4]	$5 \to 3$	I, II
TDXRay [42]	$1, 2 \to 1$	I	Cycle detection [43]	$5 \to 4, 6$	IV
ClawGuard [13]	$1, 5 \to 1$	IV	AgentProv [44]	$5, 6 \to 6$	II
Cache contention [45]	$2 \to 4$	I	Lidar [46]	$5, 6 \to 6$	II
CachePrune [47]	$2 \to 4$	I	ProcGrep [48]	$5, 6 \to 4, 6$	II
Early Bird [3]	$2 \to 4$	I	SIGLEAK [49]	$5, 6 \to 4, 6$	I
InputSnatch [50]	$2 \to 4$	I	Campaign attribution [51]	$5, 6, 8 \to 6, 8$	II
KVGov [52]	$2 \to 2, 4$	I	Dangol workflow [14]	$5, 6, 8 \to 3, 6, 8$	I
Leaky Language Models [53]	$2 \to 4$	I	AgentTell [54]	$5, 7 \to 7$	I
LLMs Have Rhythm [55]	$2 \to 3$	II	AHB [12]	$5, 7 \to 7$	III
ML privacy channels [56]	$2 \to 4$	I	FP-Agent [57]	$5, 7 \to 7$	II
MoE prompt stealing [58]	$2 \to 4$	I	PersonaFingerprint [59]	$5, 7 \to 7$	II
NetEcho [60]	$2 \to 3$	I	UI Traces [5]	$5, 7 \to 3$	I
Prompt Echo [61]	$2 \to 3$	I	Whose Agent? [62]	$5, 7 \to 7$	II
Prompt-cache audit [63]	$2 \to 4$	I	WRA [6]	$5, 7 \to 7$	II
				$5, 7 \to 3$	I

G denotes signal-generation layers and O the receiving observation layers. Layer indices: 1 hardware; 2 serving/runtime; 3 transport; 4 application/API; 5 orchestration; 6 tools; 7 browser/environment; 8 multi-agent communication. Types: I confidentiality/privacy; II attribution; III human–agent distinction/evasion; IV authorized diagnostics. These are objectives and observer roles, not mutually exclusive security guarantees. Privileged operator receivers and physical receivers are marked by the layer they inspect. Source versions, locators and the normalization of prose-only layer descriptions are in the artifact.

clean privacy improvement (one concurrent decoy reduces functional-intent reconstruction E_{func} from 0.770 to 0.610), whereas AHB reports severe task collapse (Trip Planning success falls from 0.75 Raw to 0.15 under combined `Online+Fake` humanization).

Record binding separates two axes. WRA protects user confidentiality (Type I) with discarded read-only retrievals; AHB evades a platform detector (Type III) with live GUI events that can navigate or submit forms. Their utility outcomes are therefore not contradictory. AHB's missing condition-specific denominators also preclude treating its rates as a matched causal effect.

Terminal-state reward does not establish dispatch-time authorization: an unconfirmed write could happen to produce the task's intended final state. For stateful tools, audit the real execution path and protected account/database, not only the terminal hash. E6 tests this distinction with actual simulator mutations.

5.3. Case 3: bind adaptive retraining and cost objects

The $D_0 \to D_2$ rebound and component-cost boundaries. A matched adaptation contrast compares an undefended-data-trained attacker D_0 and a defended-data-trained, defense-aware attacker D_2 on identical defended tests. UI Traces' timestamp-delay refitting and Dangol's stored/live distinction show why receiver and run provenance must accompany the score [5], [14]. E5/E6 supply direct contrasts where D_0 reaches chance and D_2 recovers the signal; E14 expands the attacker families.

On the cost side, record binding prevents cross-mechanism price substitution: Early Bird's 6ms/4% GPT-Cache semantic de-identification overhead does not price K-prefix KV-cache isolation [3]; KVGov's $1.3\,\mu$s HMAC key-derivation microcost omits workload-dependent lost cache reuse [52]; and SpliceDefense's $<$ 5ms/$<$ 1% QCP padding overhead prices chunk padding rather than combined CTBF+QCP deployment [27]. Conversely, CachePrune jointly binds marked-token attack suppression, QA/summarization utility and serving throughput on selective KV sharing [47], and OTRO bounds conditional access-count leakage ($3.94 \to 0.04$ bits given byte length) alongside separately scoped unpadded TTFT (4.59%) and epoch-padding (2.8ms) costs [16].

6. Comparability Audit Across All 58 Core Families

6.1. Unified 58-core disposition and 39-record ledger

The corpus partitions as $82 = 58$ core $+24$ adjacent families. Within the core, $58 = 20$ mitigation $+6$ contextual-record $+32$ map-only families; the result ledger contains $39 = 33 + 6$ records, respectively, from the first two groups. The six are SIGLEAK, AgentTell, FP-Agent, Behind Bars, LLMs Have Rhythm and ML privacy channels [49], [54], [57], [18], [55], [56]. They supply fixed-guardrail, access-control, interference, VPN or filter-origin contrasts rather than the audited mitigation claim. The remaining 32 comprise 22 without a scoped mitigation evaluation, five diagnostics, four unbound assertions and SplitDecoy's boundary case. S18 maps every family and record. Table 7 groups mechanisms; Table 8 reports the 20-family audit.

6.2. Within-family and within-cohort record comparisons

While the machine-readable audit retains all $\binom{33}{2} = 528$ baseline pairs for completeness, 426 pairs cross unrelated mechanism cohorts and are outside scope by construction. The 13 within-family pairs are included in the 102 within-cohort pairs: $528 = 426 + 102$, with no additional subtraction of 13. Focusing on the **13 within-family pairs** (multiple records inside the same paper) and **102 within-cohort related pairs** (66 traffic-envelope, 28 cache-service, 3 behavioral-humanization, 3 generator-fingerprint, 1 local-memory, 1 fixed-instruction; Supplement S8) reveals why paper-level comparisons fail:

- Even within the **13 same-family pairs** (plus 2 same-family pairs in the additions: `CACHEPRUNE-selective` vs. `CACHEPRUNE-context` and `OTRO-unpadded` vs. `OTRO-epochpad`), nearly half (6/13 baseline; 7/15 full) measure different primitives, targets or cost objects (e.g., Early Bird's K-prefix oracle simulation vs. its GPTCache semantic prototype; SpliceDefense's CTBF signal suppression vs. QCP-only prefill cost; OTRO's unpadded TTFT vs. derived epoch-padding cost).
- Among the seven aligned-outcome same-family pairs, four evaluate different collection or execution populations (e.g., Dangol's stored-capture feature ablation at 0.17 accuracy vs. live loopback OHTTP at 0.30 accuracy) and three lack a verified matched-test binding, so none licenses cross-record subtraction as a single paired deployment effect.
- Across the **102 related-cohort pairs**, only 16 (15.7%) support a source-qualified descriptive comparison, whereas 80 (78.4%) exhibit positively established receiver, target or cost-object mismatches and 6 (5.9%) remain unresolved (the supplied 50-pair subsample, not all 102 pairs, has 94.0% agreement across its 150 judgments, $\kappa = 0.916$).

6.3. Nine-item split reporting audit across 20 mitigation families

Table 8 reports all 180 family/item bindings. R1–R9 specify the evidence needed for scoped reuse claims, separately from deployment Checks 1–3. R4 records protected-condition fitting; R5 records a matched clean-versus-adapted contrast on defended tests. R7–R9 record inference, utility and cost. A family-level Bound entry means at least one scoped result meets the item, not that every result in that family does.

For Dangol, R5 is Bound for in-fold stored-capture experiments; the live-refit paired comparison in Table 5 remains NR-V. These are different records, so we retain the family count of two and explicitly restrict its meaning.

SoK Takeaway 2 (Joint evaluation gap). Across 20 mitigation families, 17 bind inference outcomes, 11 bind cost, 3 bind utility and 2 bind a matched adaptation contrast. Excluding structurally inapplicable entries, the latter three rates are 11/19, 3/17 and 2/11. NR-V means unverified binding, not an absent experiment. CachePrune binds inference, utility and cost at one operating point; this is not a claim that it also binds adaptive evaluation.

7. Match Transfer Claims to the Generator

7.1. Different tests rule out different explanations

A fingerprint is generated by a model interacting with tasks, prompts, tools, browser state and serving infrastructure. A holdout establishes portability along the axis it changes: new tasks, browser configurations, network topologies and hardware are distinct deployment tests. Supplement S2 maps each dependency to its supporting holdout.

UI Traces fixes a browser/action harness while testing tasks and environments. FP-Agent combines behavior with static browser attributes; shared visitor proxies are not authenticated independent people or hosts [5], [57]. Task separation can remove memorization of those task instances while retaining configuration identifiers. This is not proof of an invalid original split; it bounds portability. The traffic benchmark separately tests generating topology, so its shift is a different test rather than a stronger score on the same estimand [36].

LeakyLMs illustrates the boundary for system-property rather than agent identity. Its deployment probe observes streaming events through an ordinary client API. Its architecture branch additionally assumes known GPU type, same-class calibration and single-GPU execution [53]. Remote architecture inference therefore does not establish hardware-agnostic inference. The joint search in its remote example ranks the true configuration 28/2068, distinct from a conditional parameter search or a near match; those outcomes must not be collapsed into universal exact recovery. Source-reported local architecture generalization and remote

TABLE 7. EIGHT INTERVENTION FAMILIES ORGANIZED BY THREAT POLARITY (TYPE I: CONFIDENTIALITY PROTECTION; TYPE II: PROVENANCE/IDENTITY ATTRIBUTION; TYPE III: BOT-DETECTION EVASION), RECEIVING VIEW AND SUPPORTED DEPLOYMENT JUDGMENT. BECAUSE OBJECTIVES AND RISK MAGNITUDES DIFFER, THESE ROWS ARE NOT A COMMON EFFICACY RANKING.

Intervention & polarity	Evaluated view, target and source	Established result and applicable boundary
Padding / batching (Type I: confidentiality)	Network packet lengths and groups; topic inference. NetEcho and Whisper Leak [60], [33].	Collected padding traces retain inference signal; packet-group batching is evaluated by replay. A protected upstream length view does not cover downstream re-emission or residual grouping. Live incremental cost remains NR-R.
Constant-rate / transport cover (Type I: confidentiality)	Network workflow fingerprint; Dangol's OHTTP testbed [14].	The co-located loopback setting yields 0.30 recovery at about $13\times$ bytes and $11\times$ latency. Transformed-capture results are separate; neither result supplies objective agent-task utility.
Concurrent query decoys (Type I: confidentiality)	Destination/domain observer; private prompt and trait inference. WRA [6].	Real/decoy research prompts execute in parallel as read-only queries whose decoy answers are discarded locally; interleaved-domain evaluation reduces one-decoy E_{func} from 0.770 to 0.610. Terminal decoy utility/cost are NR-R.
Action humanization (Type III: bot-evasion)	Mobile GUI touch traces; evading human–agent bot detection. AHB [12].	Unlike WRA's read-only privacy decoys, AHB's `Fake` mode dispatches live mobile GUI touches to evade platform bot detectors, dropping Trip Planning success from 0.75 to 0.15. Non-Fake `History+Long` (0.7143 success) avoids GUI mutation.
Cache isolation / keying (Type I: confidentiality)	Co-tenant prefix timing; SafeKV, KVGov and CachePrune [15], [52], [47].	SafeKV reports post-trigger AUC near 0.5 and separate TTFT/throughput comparisons. KVGov's HMAC median 1.3 μs prices key computation, excluding lost reuse. CachePrune jointly binds marked-token protection, QA/summarization utility and serving performance.
Cache timing equalization (Type I: confidentiality)	Co-tenant cache-fusion timing; SpliceDefense [27].	Constant-Time Boundary Fusion (CTBF) suppresses the tested signal over 15 tasks/75 steps. Quantized Chunk Padding (QCP) adds mean 7.5 tokens and < 5ms/$< 1\%$ prefill time; this does not price combined deployment.
Timestamp perturbation (Type II: attribution)	Recorded website action metadata; model/agent attribution. UI Traces [5].	Fitting on delayed traces (D_2) recovers substantial attribution performance under offline timestamp edits that retain action types/order. Live execution delay and task outcomes are not measured by the trace transformation.
Oblivious / unshared memory (Type I: confidentiality)	Host or local resource observations; TDXRay, Gao et al. and OTRO [42], [23], [16].	TDXRay measures oblivious-map lookup cost; Gao measures loading/memory cost after disabling zero-copy. OTRO reduces conditional access-count information (3.94 to 0.04 bits) with separate unpadded TTFT (4.59%) and epoch-padding (2.8ms) bounds.

Authorized operator diagnostics (Type IV: ClawGuard, cycle detection, LumiMAS, LLM-serving RCA and MIRAGE) are baseline telemetry rather than privacy attacks or evasion defenses, and are cataloged separately.

deployment signatures remain positive evidence under their own assumptions.

7.2. Representation increment is not portability

Within a fixed observer and test population, coarse counts versus sequence features test whether the latter improve a specified predictor. WRA already uses ordered domains before adding timing; Kang changes class sets across some conditions [6], [62]. These test different contrasts from an order-only effect. The supplementary archive example fixes labels and units and finds a 0.3447-nat sequence increment under archive-receiver ordering. Its small common-task control leaves task-distribution explanations unresolved (Supplement S4).

Adjacent coding-agent PR attribution depends on repositories, submission pathways and artifact generation. Its mixed-feature result does not isolate an auxiliary-only C3 view; inspected code computes weighted F1 and reduces features before cross-validation, but the exact paper-run binding remains unverified [7]. We neither rewrite its headline score nor quantify optimism. WildAuth is adjacent under the same rule; input/partition ambiguities do not supply a model-independent response-only privacy claim [8].

7.3. Resulting claim hierarchy

Task-specific identification, task-disjoint identification in a fixed harness, configuration/topology transfer and hardware-calibrated architecture inference require different evidence. Move between them by changing the relevant generator while preserving target labels and the receiver contract. For a representation increment, compare representations on identical units and disclose predictor/selection restrictions. This claim-to-holdout rule lets a reader reuse each author test for the shift it measures.

8. Controlled Transfer Tests

The experiments connect observation, execution and matched evaluation. We retain artifact identifiers: E1/E3 cover buffered transport and adaptive fitting (§8.1; S11), E2 tests state-preserving cover (§8.2; S9/S11), E5–E7 isolate streaming and dispatch mechanisms, E8 tests stand-in planners (S13), E9–E12 analyze real-agent trajectories, E13 measures live tool timing/full tunnels, E14 strengthens attacks and compares classical WF adaptations, and E15 adds same-provider model trajectories. S18 gives the full concordance, including unused E4. We use *undefended* for transport without a program defense, and *standard prompt* for the prompt arm called `baseline` in immutable data. Artifact

TABLE 8. PER-FAMILY EVIDENCE BINDINGS FOR ALL 20 EVALUATED-MITIGATION FAMILIES. R1–R9 ARE SCOPED REUSE ITEMS, NOT OVERALL PAPER SCORES.

Family	R1	R2	R3	R4	R5	R6	R7	R8	R9
UI Traces [5]	**B**	**P**	**B**	**B**	**B**	**B**	**B**	U	–
WRA [6]	**B**	**P**	**B**	–	–	**P**	**B**	**P**	U
NetEcho [60]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	U
Early Bird [3]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	**B**
Whisper Leak [33]	**B**	**B**	**P**	?	?	**B**	**B**	U	**B**
SpliceLeak [27]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
Time Will Tell [31]	**B**	**B**	**B**	?	?	**B**	**B**	U	**B**
I Know What You Said [23]	**B**	**B**	**B**	–	–	**B**	U	–	**B**
Speculative decoding [25]	**B**	**B**	**P**	?	?	**B**	**B**	–	**B**
Dangol workflow [14]	**B**	**B**	**B**	**B**	**B**	**B**	**B**	U	**B**
Cloud traffic benchmark [36]	**B**	**P**	**B**	?	?	**P**	**B**	**P**	U
Remote timing [20]	**B**	**B**	**B**	–	–	**B**	**B**	U	**P**
TDXRay [42]	**B**	**B**	**B**	–	–	**B**	U	**B**	**B**
SafeKV [15]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
AHB [12]	**B**	**B**	**B**	**P**	?	**B**	**B**	**B**	U
KVGov [52]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
CachePrune [47]	**B**	**B**	**B**	–	–	**B**	**B**	**B**	**B**
OTRO [16]	**B**	**B**	**B**	–	–	**B**	**B**	–	**P**
SparLeak [39]	**B**	**B**	**B**	?	?	**B**	**B**	**P**	U
Shape and Substance [37]	**B**	**B**	**P**	U	U	**B**	**P**	U	**P**
Bound	20	17	17	4	2	18	17	3	11
Applicable	20	20	20	11	11	20	20	17	19

B: bound; **P**: partial; U: unreported in inspected scope; ?: unverified binding (11 cells); –: inapplicable. R1 target/polarity; R2 receiver; R3 placement/execution; R4 defended-condition fit; R5 matched adaptation; R6 workload/generator shifts; R7 inference; R8 utility; R9 cost. A bound cell is supported by at least one scoped record. Dangol R5 concerns stored capture; its live-refit binding remains unverified. CachePrune jointly binds R7–R9 at one operating point. Source follow-up resolves three former ? cells: cloud-traffic R3 to B; Shape and Substance R4/R5 to U (S8).

keys `raw`/`baseline` are retained for reproducibility. Seven author-constructed mechanism checks were recorded before E5–E7 (S12); they are engineering checks rather than a sample of external review decisions.

For E6/E7/E9, a cell meets the *evaluated-suite criterion* when maximum BA has an upper 95% bound below 0.60 and maximum AUROC is below 0.65. This absolute-threshold assessment is separate from the measured reduction relative to undefended transport: a lower BA bound above 0.60 indicates above-threshold leakage, while an interval crossing 0.60 leaves the absolute threshold inconclusive. We recompute the suite maximum within each of 2,000 common bootstrap draws: tasks for E6/E7 and customers for E9, retaining their tasks and repeats together. Intervals condition on fitted predictions and do not cover unknown attackers.

A byte-transparent relay captures post-handshake TLS application records on loopback connections. TLS 1.3 session tickets are disabled. E5–E8 use saved text or stand-in planners; E9 invokes a real model and executes its structured actions in the local public retail simulator. No production account or tool is modified.

TABLE 9. DENOMINATORS AND UNDEFENDED NETWORK BA (CANCEL/RETURN, %). EPISODES INCLUDE BOTH PROMPTS; NUMERICAL PAIRS USE THE STANDARD PROMPT. THESE ARE DIFFERENT ESTIMANDS.

Study	Observation / fit	Episodes	BA pair
E9	Replay lengths/order; original frozen suite	276 (138)	97.2/94.0
E10	Analytic lengths/order; 11-attack cross-fit	690 (345)	91.7/85.5
E13/14	New RPC envelope; lengths/order + time	276 (138)	91.7/86.4

Parentheses: episodes per prompt. E9/E13 use 24 test customers; E10 pools 53. E14 expands E13's attackers. E13 bytes use its own undefended envelope, not E9's serialization.

TABLE 10. E5: THE SAME 48 SAVED RESPONSES EMITTED BY A BUFFERED OR A TOKEN-STREAMING FRONTEND (TLS 1.3 LOOPBACK CAPTURES). BYTES ARE RELATIVE TO THE UNPADDED EMITTER OF THE SAME KIND. TTFB/DONE: FIRST AND LAST DELIVERED BYTE IN TOKEN INTERVALS (ACCOUNTING PROXY). ACCURACIES ON 24 HELD-OUT OUTPUTS (12 TOPICS); ID: CLOSED-WORLD TOP-1 IDENTIFICATION AMONG 48 RESPONSES (CHANCE 2.1%).

				T1 acc.		T3	
Emitter / framing	Bytes×	TTFB	Done	D_0	D_2	D_2	ID
Buffered, unpadded	1.00	105	105	100.0	100.0	91.7	87.5
Buffered, training cap	2.19	105	105	50.0	50.0	50.0	2.1
Stream, per token	1.00	1	105	100.0	100.0	87.5	100.0
+ random event pad	1.29	1	105	100.0	100.0	87.9	62.3
+ constant event size	1.23	1	105	100.0	100.0	87.5	58.3
8-token batches, const.	0.21	8	105	50.0	100.0	87.5	20.8
Const. events + count cap	2.42	1	208	50.0	50.0	50.0	6.3

D_2 is the best of total-byte threshold, record-count threshold and a logistic sequence model, each fitted on defended training outputs. Random padding: 10 socket-measured draws plus an independent profiling run. TLS 1.2 gives identical accuracies and byte ratios within 0.03. Tokens: o200k_base proxy. Count cap: 208 (T1) and 88 (T3) events fitted on training outputs; 2/24 and 5/24 test outputs overflow it.

8.1. Transport: buffered and streaming emitters

The buffered tests use brief/expanded saved responses for 24 topics (12 training, 12 test). A two-link TLS replay remains 100% separable on the terminal link when only upstream traffic is padded; padding both reduces verbosity inference to 50%, reproducing NetEcho's mechanism [60]. A training-fitted cap reaches 50% at 2.18× bytes. Defended-data fitting recovers 13.5–50.0 points under offsets, buckets and random tails (S11).

E5 emits the same 48 texts as per-token server-sent events over TLS 1.2/1.3, reflecting the streaming surface [1], [33]. Per-event padding preserves token count: verbosity remains 100% separable. Eight-token batching cuts bytes to 0.21× and clean-trained BA to 50%, but refitting recovers 100%. Constant shape reaches 50% at 2.42× bytes and 208 versus 105 token intervals; two overflowing test outputs remain identifiable. Table 10 gives all emitter, attacker and cost cells. A buffered cap delays the first byte until generation ends, so its cost cannot price streaming protection.

TABLE 11. E6 MAIN DISPATCH RESULTS: ALL 115 REFERENCE PROGRAMS, THREE SEEDS, KNOWN-POPULATION DEFENSE CALIBRATION; TASK-FOLDED ATTACKERS. MAXIMUM BA (CANCEL/RETURN); EXTRA CALLS PER REAL CALL; DATABASE EQUALITY.

Policy	BA (%)	Calls+	DB eq.
No cover	100.0/100.0	0.00	100%
Cancel, $p = .5$	86.1/100.0	.10	100%
Cancel, $p = 1$, shadow	100.0/100.0	.20	100%
Cancel, $p = 1$, live	100.0/100.0	.20	0%
Cancel count $\rightarrow 2$	76.9/100.0	.35	100%
One decoy program	91.9/85.9	1.01	100%
Two decoy programs	83.5/78.1	1.98	100%
Shape equalization	50.0/54.5	6.71	100%

8.2. Execution: state scope and randomized cover

State scope. In the 115 public retail programs, same-item modification changes status and appends a refund even when item and price are unchanged (173/291 pre-insertion units). State preservation $f(s, a) = s$ therefore differs from idempotence $f(f(s, a), a) = f(s, a)$. Cancelling another user's order changes the shared database. Shadow cover separates an observable call from its live effects: authorized reads, noncommitting API operations or a trusted gateway's isolated executor can supply this boundary (§9.1). A known-position cover call remains removable with 100% accuracy (S11).

E6: randomized cover against policy-aware attackers. E6's dispatch observer sees ordered tool names and response lengths. Cover is sampled from other users' programs, randomly interleaved and executed on shadow state, with live execution as a control. Policies add cancellation, pad its count, interleave decoy programs or equalize every tool's count. Templates, donors and caps use all 115 reference programs: this is known-population defense calibration. Attackers use clean/defended gradient boosting and logistic regression on counts, bigrams and lengths, plus a count rule. Task-grouped 5-fold evaluation uses three seeds and cancellation/return targets (18/32 positive tasks).

Table 11 reports all dispatch policies. One cancellation drives undefended-data-trained BA to 50%, but refitting recovers 100% (P2; full table in S12). Count equalization defeats the count rule while correlated tools yield 76.9% (P4). Decoy programs, the read-only pattern of WRA [6], leave 83.5–91.9%. Shape equalization reaches 50.0%/54.5% at 6.7 extra calls per real call. Database equality is 100% for shadow execution and 0% for live cancellation, demonstrating the importance of the execution boundary.

8.3. E7: composing execution and transport on the agent–tool link

E7 composes dispatch and transport on an encrypted agent–tool link (Figure 1). Four cover policies yield 4,140 TLS 1.2 sessions. Framing is undefended, 1024-byte buckets or constant requests/responses with overflow buckets. The evaluated network projection contains direction-tagged TLS record lengths and order; it excludes tool names and inter-arrival times. A profiling attacker learns size-to-tool mappings from training-fold API calls and applies dispatch attacks to decoded traces; S12 specifies caps and classifiers.

Request-only tool identification reaches 89.9%/91.4% under cancellation/return folds (582 calls; majority 29.4%). Count and decoy cover leave 72.7–95.7% network BA, extending their dispatch leakage (P5). Framing alone leaves bucketed return BA at 73.9%. Shape cover gives 51.2%/60.5% with undefended framing and 54.4%/50.0% with constant framing. No configuration meets the absolute criterion for both targets. Joint constant framing costs 26.3× bytes and 7.7× calls with 100% database equality: one operating point, not a minimum cost (P7). E8's stand-in planners (S13) motivate the causal, training-only dispatcher evaluated next with real model decisions.

Transport link versus dispatch boundary. Per-call framing leaves call counts and turn structure exposed. A full traffic-flow-confidentiality tunnel can instead conceal these from a passive link observer using fixed-size, constant-rate traffic [72]; protecting session duration additionally requires a workload-independent traffic horizon. Its dummy packets need not execute tools. E7/E9 test per-call framing; E13 separately measures a complete fixed-rate TLS tunnel. At a dispatch boundary that exposes calls after decapsulation, transport padding is removed and call-level cover becomes relevant. Protection then depends on hiding cover markers and downstream routing from that observer (S17).

8.4. E9: real agents and training-only defense calibration

E9 tests defense reuse on autonomous model-generated trajectories under a strict calibration boundary. GPT-6.1 Sol (low reasoning effort) plans and responds from the retail policy, public tool schemas and actual sandbox observations. A separate context of the same model simulates the user. The rules arm adds one customer-details read before order access and a read before each order write. All 115 tasks have two prompt arms and three repeats, completing the frozen 690-run protocol. All 414 training runs on 69 tasks precede freezing of a shared count template, donor pools and framing caps; the 276 test runs cover 46 tasks from 24 disjoint customers.

Planner-visible equivalence. Let H be the planner's message/tool-observation history, excluding wall-clock timing. Real calls and responses pass unchanged; cover uses isolated state and hidden responses. Thus the planner projection satisfies $\pi_{\mathrm{plan}}(D(H)) = H$ (Appendix A). E9 replays each realized trajectory through four program conditions, isolating the program-layer effect from LLM sampling variation. Online denotes prefix-only scheduling; this equivalence does not preserve elapsed time, queueing or timeout behavior. The trusted cover executor knows which calls are shadow calls; E9 protects the measured dispatch/network projections, not this executor or a provider's commit log.

The dispatcher inserts training-derived cover using only past calls and flushes unmet counts at episode end. A

timing-aware observer could exploit this flush or read/write service times; E9 measures the length/order projection, not a defense against that observer. Four program conditions per prompt isolate undefended transport, padding, shape cover and their combination. Training-fitted attackers evaluate dispatch and network observations across 2,760 TLS replay sessions. Every test task, failure and handoff remains in the analysis. Supplement S14 supplies the full ablations and customer-cluster intervals.

Behavioral control and frozen-split reference. Table 12 retains the original E9 result and calibration boundary. Rules raise mechanical read-order adherence from 86.2% to 97.1%, while undefended intent BA remains above 93%. This is a control showing that the imposed read constraints do not remove the write-associated signal; the prompt was not designed as a privacy defense. Task reward is 110/138 under the standard prompt and 105/138 under rules (paired difference -3.6 points, interval $[-9.5, 1.4]$). Program-condition reward is inherited from reused logical trajectories, not remeasured planner utility.

The frozen joint-shaping cell has chance-level point estimates, with upper bounds 60.6%/65.5%; the absolute criterion is unconfirmed. The paired decreases are 47.2/44.0 points. Rules plus joint shaping leaves return BA at 60.6%, demonstrating sensitivity to the prompt/cohort realization. Five cancellation-positive and seven return-positive test customers limit precision. We therefore use the all-customer cross-fitted comparison below as the main descriptive summary, retaining E9 as the originally frozen reference rather than relabeling it an independent confirmation.

8.5. E10–E12: cross-fitting, operating points and mechanism

E10: primary within-cohort summary (post hoc). A post hoc customer-grouped stratified 5-fold analysis reuses all 690 runs: 53 customers, including 14 cancellation-positive and 18 return-positive customers. Each fold fits defense templates, donors, caps and attackers on the other customers only. Its analytic TLS 1.2 model matches all 2,760 measured sessions and 120 fresh captures. We expand all nine prior attacks with a trained 1D-CNN and k-FP-style forest-leaf attack; the primary table uses the maximum over all 11. Undefended BA is 91.7%/85.5%. Recalibrating the joint shadow-count/constant-framing design reduces BA by 31.2/30.0 points (paired 95% intervals [15.3,41.0]/[14.8,39.2]); joint buckets reduce it by 34.6/32.8 points. Stronger learners preserve these reductions while increasing some defended estimates. Both joint designs leave the absolute criterion inconclusive (S21).

E11: choosing the operating point. The historical E11 selection uses the original suite and six configurations. Choosing the cheapest point with all four inner-fold BAs $\leq 60\%$, using only the original 69 training tasks, picks 1024-byte buckets. On the 46-task test split, byte cost falls from 15.4× to 9.8×. S15 retains its original predictions and intervals. This retrospective selection illustrates pricing under an explicit rule; it is not an independent confirmation or a certification under the stronger suite. Table 13 supplies the updated full-cohort comparison.

E12: locating the signal. The expanded 11-attack oracle ablation (Table 14) retains 91.7%/84.9% BA in state-changing exchanges, which comprise only 19.6% of real tool calls. Removing them reduces BA to 62.8%/59.9%: paired decreases of 28.9 [10.7,41.9] and 25.6 [11.0,35.4] points. Writes therefore carry the dominant measured signal, while reads retain residual information. Rules reduce distinct read sequences from 74 to 62, while write sequences remain 37 versus 36. The oracle selects exchanges using true tool types, then exposes only their network lengths/order; it is a localization diagnostic, not a deployable filter.

8.6. E13: live timing, latency and a complete tunnel

E13 executes all 690 saved agent action trajectories through a live local tool server under six conditions, yielding 4,140 new TLS sessions. Each real response gates the next action; the server executes real and shadow calls on separate states. The original customer-disjoint 414/276 training/test trajectory split is retained. Attackers are fitted and frozen on training captures before test collection. The relay records ciphertext lengths, directions and arrival times. These clocks measure the tool RPC chain, excluding LLM generation and WAN transit (S19).

The complete tunnel sends 4,096-byte cells in both directions at 2 or 5 ms target intervals, including dummy cells. Its training-sized 133-slot horizon remains fixed after the last response; dummy cells execute no tools. Serial and two-in-flight shadow conditions use E9's prefix-only count policy and constant per-call framing. The full attack suite unions size/order, timing and joint features, plus learned flush removal for shadow cover; its detectors and transformations use training customers only.

Measure the exposed clock. Under the original E13 suite on standard-prompt test trajectories, serial shadow cover yields 50.0/50.7% size/order BA and 61.4/53.4% under its full suite. Its cancellation increase is 11.4 points (95% paired interval [3.7,19.6]), identifying a residual clock signal. With two in-flight calls, both views yield 51.7/57.7%. Across both prompt arms, learned flush detectors reach 90.4%/92.3% BA for serial/parallel scheduling (S19).

Price the protected boundary. In that original suite, the 2 ms tunnel yields 62.5/52.9% full-suite BA and the 5 ms tunnel 53.1/57.3%, with one tool call per original action at 125.7×/125.7× bytes. Serial versus two-in-flight shadow scheduling has real-chain p95 latency of 26.2/20.8 ms; the fast/slow tunnels have 56.4/112.3 ms. All 4,140 sessions preserve every real response and final live state. The tunnel has fixed cell counts and sizes, but observed direction/order and host scheduling remain measurable; neither tunnel nor shadow schedule meets the two-target absolute criterion. The expanded-suite Table 15 retains the same costs and strengthens the attack maximum. Transport cover avoids tool execution on the passive link; call-profile cover addresses dispatch beyond tunnel termination, at substantial extra cost.

TABLE 12. E9: COMPLETE REAL-AGENT EVALUATION WITH TRAINING-ONLY DEFENSE CALIBRATION. TESTING COVERS 46 TASKS FROM 24 HELD-OUT CUSTOMERS, TWO PROMPTS AND THREE REPEATS (276 EPISODES). NETWORK BA IS THE MAXIMUM OVER THE FIXED ATTACK SUITE, WITH CUSTOMER-CLUSTER 95% INTERVALS.

Condition	Cancel BA [95% CI]	Return BA [95% CI]	Reward (%)	Bytes×	Calls×
Undefended	97.2 [90.0, 100.0]	94.0 [85.8, 100.0]	79.7	1.00	1.00
Rules prompt	97.2 [90.0, 100.0]	93.5 [85.0, 99.5]	76.1	1.00	1.00
Padding only	66.1 [50.0, 83.9]	55.5 [50.0, 73.0]	79.7	4.02	1.00
Shape only	60.8 [55.4, 71.3]	51.9 [50.5, 57.8]	79.7	4.16	3.83
Joint program	50.0 [50.0, 60.6]	50.0 [50.0, 65.5]	79.7	15.39	3.83
Rules + joint	50.0 [50.0, 65.1]	60.6 [50.0, 73.3]	76.1	15.72	3.88

Paired joint-minus-undefended BA differences (percentage points): standard-prompt cancellation -47.2 [-50.0, -33.3], return -44.0 [-48.9, -27.8]. These post hoc contrasts use the same customer draws and reselect the attack-suite maximum in each condition. Joint = online shadow cover plus constant framing; padding/shape ablations use standard-prompt trajectories. Ratios use undefended transport within each prompt. Shadow isolation preserves the logical planner transcript; the four program conditions replay actual LLM trajectories over local TLS 1.2. Full methods, intervals and prompt contrasts: Supplement S14.

TABLE 13. PRIMARY E10 CROSS-FIT SUMMARY UNDER THE EXPANDED 11-ATTACKER UNION: ALL 690 EPISODES AND 53 CUSTOMERS. MAXIMUM NETWORK BA WITH CUSTOMER-BOOTSTRAP 95% INTERVALS. STANDARD PROMPT UNLESS MARKED RULES; BYTE/CALL RATIOS RELATIVE TO UNDEFENDED TRANSPORT. JOINT-DESIGN ABSOLUTE-CRITERION OUTCOMES REMAIN INCONCLUSIVE.

Configuration	Cancel BA (%)	Return BA (%)	B×/C×
Undefended	91.7 [82.1, 100.0]	85.5 [76.7, 93.4]	1.0/1.0
Joint constant	60.5 [52.9, 72.4]	55.5 [51.8, 64.2]	19.8/5.0
Joint buckets	57.1 [50.0, 71.5]	52.7 [50.0, 61.7]	12.4/5.0
Rules + buckets	56.9 [50.2, 71.2]	52.7 [50.0, 65.0]	12.4/5.0

Joint constant is E9's shadow-count plus constant-framing design, refit within each fold. Joint buckets changes framing to 1024-byte buckets. Every bootstrap draw reselects the complete maximum; S21 reports both prompts, AUROC and paired reductions.

TABLE 14. E12 ORACLE LOCALIZATION UNDER THE 11-ATTACK UNION. STANDARD PROMPT: 345 CROSS-FITTED EPISODES, 53 CUSTOMERS. MAXIMUM NETWORK BA (%) WITH CUSTOMER-BOOTSTRAP INTERVALS.

Target	Full trace	Writes removed	Writes only
Cancel	91.7 [82.1, 100.0]	62.8 [54.6, 75.3]	91.7 [82.1, 100.0]
Return	85.5 [76.7, 93.4]	59.9 [52.2, 71.8]	84.9 [78.8, 94.0]

8.7. E14: stronger D2 attacks and classical controls

We fit seven R7 D2 models, including a trained 1D-CNN, k-FP-style leaf-fingerprint nearest neighbors and tabular learners, while retaining every original E13 attack. All fit on defended training customers under the public algorithm/configuration. The expanded maximum increases some tunnel scores; it preserves the serial-shadow clock result. Every bootstrap draw reselects the full maximum.

Table 15 also adapts Tamaraw, FRONT, WTF-PAD and Walkie-Talkie to the same undefended captures. These are offline RPC adaptations: fixed cells preserve input-derived fragment sizes, directions and order, and dummy cells execute no tools. Random-donor WT reaches 50.0/56.7% BA at 2.04× bytes; fast Tamaraw reaches 57.8/51.8% at 11.99×. Neither supports an absolute-security claim. WT lacks symmetric sensitive/non-sensitive pairing; the WTF-PAD adapter uses training-IAT histograms rather than the original Tor configuration. The short-customer FRONT calibration tests distribution mismatch without dropping test customers. S20 provides both prompts, algorithms and a cost plot under the same seven common attacks, alongside the larger available unions. Byte ratios share a denominator; measured RPC latency and simulated schedule span have distinct origins. These points expose cost/leakage trade-offs, not uniform shadow-cover superiority.

8.8. E15: an additional model on the same environment

A separate 150-run GPT-6 Luna/low study uses 75 existing retail tasks, two prompts and one repetition; ten pilots are excluded. Under the standard prompt, a three-model dispatch diagnostic gives writes-only BA of 87.5/66.6%, versus 50.0/57.0% for get/find/list reads. The rules arm likewise retains more measured signal in writes. Task rewards are 32/75 and 33/75 for standard/rules; all failures and seven protocol-error episodes remain included. Each prompt-specific attacker fits only 29 training episodes, and the inherited gradient booster cannot split at this sample size. This smaller-suite, same-provider check supports the write-associated pattern without establishing cross-environment transfer. Padding alone matches or improves the joint network point estimates here. S22 gives every cell, actual elapsed times, the distinct read-view definition and 600 local TLS replays with training-only calibration.

8.9. What the cases establish

The experiments test the comparison obligations. E5 ties protection to the emitter; E6/E7 tie cover to state and the final network view; E9–E12 locate the write-associated signal and price training-calibrated shaping. E13 exposes the clock and tool-response cost. E14 tests stronger learners and established WF mechanisms on the same saved workload; E15 checks a separate model alias. The supported outcome

TABLE 15. E13 CAPTURES WITH THE E14 EXPANDED DEFENSE-AWARE SUITE, AND OFFLINE WF ADAPTATIONS. STANDARD PROMPT: 138 TEST TRAJECTORIES, 24 CUSTOMERS. BA AND UPPER 95% BOUNDS ARE CANCEL/RETURN PERCENTAGES. EVERY CONDITION HAS AN UPPER BOUND ABOVE 60%, LEAVING THE TWO-TARGET ABSOLUTE CRITERION UNCONFIRMED. LAST-REAL/FINISH P95 IS MEASURED FROM RPC START FOR M ROWS; FOR S ROWS IT IS SIMULATED FROM THE FIRST INPUT RECORD, NOT ONLINE PLANNER LATENCY.

View	Condition	BA	Upper BA	Bytes×	Calls×	Last-real p95 (ms)	Finish p95 (ms)
M	Undefended	91.7/86.4	100.0/97.2	1.00	1.00	15.4	15.5
M	Per-call constant	68.6/62.7	92.8/79.5	3.92	1.00	16.6	16.7
M	Serial shadow	61.4/53.4	71.4/65.8	15.11	3.83	26.2	46.6
M	Two-in-flight shadow	51.7/57.7	69.1/69.7	15.11	3.83	20.8	37.4
M	Full tunnel, 2 ms	62.5/55.2	70.2/62.0	125.71	1.00	56.4	270.4
M	Full tunnel, 5 ms	54.6/57.3	72.9/64.2	125.71	1.00	112.3	666.6
S	Record buckets	53.1/69.8	78.5/86.9	2.25	1.00	13.8	13.8
S	Tamaraw 40/12 ms	55.3/55.3	68.7/64.4	12.15	1.00	648.0	8000.0
S	Tamaraw 4/1.2 ms	57.8/51.8	71.8/61.8	11.99	1.00	70.8	800.0
S	FRONT	59.0/69.3	69.9/81.8	2.51	1.00	13.8	13.8
S	WTF-PAD adaptation	51.9/62.7	64.1/79.6	1.88	1.00	13.8	13.8
S	WT random-donor	50.0/56.7	67.0/67.8	2.04	1.00	16.6	16.6
S	FRONT, short calibration	52.4/56.0	61.3/71.1	2.33	1.00	13.8	13.8

M rows retain 17–22 attacks (seven R7 models plus all original E13 attacks); S rows use the common seven. These strongest-available maxima are not a matched-suite dominance comparison. S20 provides the common-seven sensitivity and cost plot. Bytes include TLS headers; ratios use each prompt's undefended captures. WF dummies execute no tools; random-donor WT has no symmetric site-pair guarantee.

TABLE 16. DEPLOYMENT BOUNDARY: THE RELEVANT PASSIVE OBSERVER AND WHAT THE MEASURED MECHANISM CAN PROTECT.

Setting	Visible information	Applicability / trusted point
Same-host sidecar	Host trace/process/IPC metadata, if permitted	Link padding alone does not hide host instrumentation; isolate the monitor's access.
Same-VPC gateway	Ingress record sizes, times and addresses	Tunnel or shadow shaping can protect this link; gateway/markers remain trusted.
Remote SaaS/MCP	IP, unprotected SNI, per-endpoint flows and times	Shared ingress needed for tool-identity cover; ECH alone does not hide destination IP.
Cross-organization	Each hop's routing; provider calls/commit log	No guarantee across all hops. A trusted isolated executor is an explicit added assumption.

is a measured trade-off under explicit observation and execution contracts, not an absolute security bound or a novel padding construction.

9. Implications, Ethics and Limits

9.1. Deployment paths and reporting obligations

Table 16 identifies four deployment settings and their observers: access-network operators, VPC monitors, relays, or dispatch observers seeing calls after decryption. Encrypted ClientHello hides the inner server name when supported, while destination IPs and exposed downstream connections can still identify tools [73]. Our single-ingress captures establish no multi-endpoint anonymity.

Shadow state has three implementation paths. **(1) Authorized reads and verified no-ops** use the live backend with responses discarded; idempotence alone does not preserve state, and read cover cannot hide exposed write types. **(2) Provider-supported isolation** includes Kubernetes server dry-run, isolated Stripe sandboxes and transactional rollback [74], [75], [76]. Dry-run validates without persistence, sandboxes maintain separate state, and rollback covers transactional effects; sequence changes can survive rollback [77]. **(3) A trusted MCP/tool gateway** decrypts authenticated cover markers and routes shadow writes to an isolated simulator. Real and cover traffic share an ingress envelope, with routing/timing/size differences treated as attack features. E13 measures a local gateway; the provider-specific paths are deployment designs.

Protection ends where an observer learns cover markers, routing branches or committed writes. The executor itself and provider commit logs are outside the ingress mechanism's protection (S17). Report leakage together with backend cost, latency, quotas and downstream observations. Cover consumes billable work and shared capacity and can amplify denial of service. Deployment requires per-tenant admission budgets, backpressure and separate sandbox credentials. Exhausted quotas must expose the loss of protection, rather than silently send shadow writes to live state.

9.2. Ethics and artifact scope

Executions use cloned public benchmark databases and loopback sockets; model requests contain public tasks and simulated interactions. No production tool or real account is modified. Stakeholders include paying tenants, providers sharing capacity and people whose records writes could affect. Isolation and bounded authorized cover are necessary for deployment. Other-account cancellation is a simulator control. No third-party live service was probed for a vulnerability, so no vendor-disclosure event is claimed. Released inference code has surveillance uses; the examples target

public/synthetic identities and controlled local endpoints. Bot-detection evasion has a different objective and authorization boundary [12], [6].

The artifact preserves source locators, model decisions, observations, frozen calibration and executable checks. AI assisted field extraction, implementation, source reconciliation and prose revision; authors verified the archived source-bound entries in §2.3. This use addresses the volume of reconciliation and supplies no independent human ratings. Submission requires an anonymous immutable snapshot with identifying metadata removed; the development Git history is not that snapshot. Saved-output replays, model-generated trajectories and local TLS measurements are labeled separately.

9.3. Threats to validity

The purposive corpus has no field-wide recall estimate. Extraction–verification agreement is non-independent; standardized blind recoding sheets support future community audit. The 24 constructed cases measure rule application, not external decision accuracy. Eleven audit cells remain unverified after source inspection and three explicit status corrections.

E5 uses 48 saved outputs and a proxy tokenizer; E6/E7 use reference programs with known-population calibration. E9–E12 use one retail environment and a mutable model alias also serving as user simulator; development had access to all 115 tasks. Cross-fitting expands customer coverage within that population, and intervals condition on fitted attackers. E10–E14 are post hoc. Program conditions preserve reward structurally through trajectory reuse. E13 measures local RPC clocks on fixed actions under Windows scheduling; LLM generation, WAN transit, timing-dependent replanning and production quotas are unmeasured. A deployed tunnel horizon must also cover planner think time. The separate Luna extension supplies a same-provider retail comparison. New independent customers, airline/MCP transfer, fixed open weights, cross-vendor models and live replanning under cover remain untested. Power depends on independent positive customers and attacker correlation; repeating old tasks adds no independent customers.

10. Conclusion

This SoK binds published defense results to their receiver, execution boundary and evaluation conditions. Separating records reveals why upstream shaping, read-only cover, matched adaptation and component costs cannot be transferred interchangeably. The real-agent analysis identifies state-changing calls as a dominant signal and measures how shaping changes inference at substantial cost. Timing-aware evaluation and classical traffic-defense comparisons further delimit that effect. The reusable contribution is the evidence framework and its mechanism-level findings: evaluate the protected observation, preserved state and cost on the same workload, and distinguish paired improvement from an unconfirmed absolute bound.

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Appendix A. Reproducibility and Planner-Visible Equivalence

Artifact organization. The archived reproducibility package groups the source ledger and family audit, transport experiments, stateful tool-dispatch experiments and real-agent studies. A machine-readable module index resolves these groups and their extensions to source files, predictions and verification entry points. The README supplies environment versions, checksums and commands. The complete 690-run cohort and earlier releases retain their original membership and results for provenance.

The source module contains all 82 family records, the 39-result ledger, all 180 audit cells, search reconciliation and coding columns. The real-agent module contains the 414/276 training/test split, frozen calibration, model decisions, sandbox scores, 2,760 TLS sessions and saved attack predictions. Offline reconstruction reproduces the fitted-attack results and intervals without additional model calls. The real-agent module also includes customer-cross-fitted analyses, training-fold operating-point selection and an oracle write-call ablation (E10–E12, Supplement S15). Their analytic TLS record model is checked against the measured sessions. E13 adds 4,140 live tool sessions, observed record-arrival timestamps, pre-test frozen attack models and complete tunnel controls (S19). The paired program contrasts reported in the current paper are a post hoc analysis of those saved predictions, using the same 2,000 customer-cluster draws and recomputing the suite maximum for each condition in each draw.

The robustness extensions archive every stronger attack, prediction and defense transform. E14 rechecks measured clock leakage and compares explicitly specified WF adaptations (S20); the primary E10 and E12 updates retain all prior attacks and add CNN/forest-leaf learners (S21). Exact local refitting, model reloading and relocated-path verification make these outputs executable without new model calls. This is computational verification, separate from external independent replication or human recoding. Source follow-up corrections are an explicit overlay on the historical audit, preserving its original columns and agreement statistics.

E15 adds a separate Luna cohort (S22). Its public raw-event archive, dependency hashes and executable checks link model decisions to sandbox rewards and local TLS replays. Recalculation uses the archived benchmark/tool sources and the public raw records without new model requests. The original Sol cohort, fitted calibration and earlier outputs remain unchanged.

Equivalence invariant and assumptions. Let s_t be live tool state and H_t the planner-visible sequence of messages, actions and tool responses. A real action a_t produces $(s_{t+1}, o_t) = T(s_t, a_t)$. The defended dispatcher forwards a_t unchanged and executes cover transitions only on a separate state s_t^{sh}, suppressing their responses. With the same initial live state, deterministic simulator tools and the same realized real actions, cover cannot change s_{t+1} or o_t. Induction on real-action steps gives

$$s_t^D = s_t, \qquad \pi_{\mathrm{plan}}(D(H_t)) = H_t.$$

Framing preserves application payloads and end-of-episode flushing adds only hidden shadow transitions. This is equality of the logical planner transcript. It assumes no timeouts, shared-state side effects or timing-dependent tool behavior; equality of independent model samples is not required. Beyond E13's local tool-loop measurements, a fully interactive deployment would additionally need to establish that wall-clock delay and shared resource effects do not change task execution.

The frozen dispatcher copies each real-call record verbatim into the defended stream. Executable checks match the real-call projection and original scoring fields across all 690 trajectories and four program conditions. Shadow execution and wire-byte accounting are separately reconstructed by the original full-cohort verifier. Reward equality therefore expresses the isolation invariant; call and byte ratios quantify the transport intervention on that fixed workload.

Reading the main tables. The corpus map reports signal-generation and receiving layers separately. The per-family audit distinguishes bound, partial, unreported, unverified and inapplicable evidence. These statuses preserve source scope: an unverified binding identifies a comparison obligation rather than declaring an experiment absent. In the real-agent table, point BA, paired BA reduction and the absolute finite-suite criterion answer different statistical questions; all intervals condition on the saved fitted attackers.